Operating Systems · 21CSC202J

Unit 1 · Session 7 · SLO 2 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 7 : SLO â€“ 2 | Protection and Security â€” Scrambled Sentences

I. Sentences in Correct Order (Protection and Security Mechanisms)

1. Protection ensures that each process accesses only the resources it is authorized to use.
2. The OS uses an access matrix of domains versus objects to define allowed operations.
3. User authentication (passwords, biometrics) verifies identity before granting access.
4. Access control lists and capability lists implement the access matrix in practice.
5. Encryption protects data confidentiality during storage and transmission.
6. Firewalls and intrusion detection systems defend the system from network threats.
7. Auditing and logging record security-relevant events for later analysis.

II. Summary

• Protection = controlling internal access to resources (access matrix, ACLs, capabilities).
• Security = defending against external/internal threats (authentication, encryption, firewalls).